A Geometric Theory of Contextual Semantic Capacity: Concentration of Measure, Superposition, and Readout Limits in Large Language Models

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High-dimensional concentration geometry supplies the kinetic capacity for contextual semantic encoding in Large Language Models: many candidate semantic directions, frames, or low-dimensional sense subspaces can coexist in a finite-dimensional activation space with small pairwise overlap. This capacity, however, does not imply unrestricted semantic access. The same residual coherences that make superposition possible accumulate when many semantic components are simultaneously active, producing an interference scale of order $\sqrt{k/d_{\text{eff}}}$, where d_{eff} is the effective dimension of the activation channel actually used for readout. We therefore distinguish *kinetic semantic capacity*—what can geometrically coexist—from *epistemic semantic accessibility*—what can be reliably read, discriminated, and composed. Drawing on a structural analogy with contextual frames, we analyze a polysemous token as a candidate hinge vector that gates access to sense-bearing directions or low-dimensional sense subspaces in its orthogonal complement. Lévy-type concentration bounds explain the abundance of possible senses; interference and sparse-recovery bounds explain why only a limited active subset can be accessed at once. The result is a geometric capacity theory for contextual semantics: LLM dimensionality acts not merely as storage space, but as a finite concurrency bandwidth for linearly accessible meaning.

Keywords: Large Language Models, Embeddings, Polysemy, Concentration of Measure, Lévy's Lemma, Contextual Readout, Hinge Vectors, Superposition, Compressed Sensing

I. INTRODUCTION

Large Language Models (LLMs) operate on dense vector representations of tokens in high-dimensional spaces. The success of these systems on linguistic tasks has been accompanied by a body of folk-theoretical claims about how meaning is encoded geometrically: directions correspond to features, distances to similarity, and arithmetic operations on vectors to semantic analogies [1]. Yet the mathematical underpinnings of these claims remain only partly articulated.

This paper addresses the following geometric question: what are the capacity limits of contextual semantic representation in a finite-dimensional activation space? LLMs must encode not only many possible semantic features, but also the context-dependent rules by which a small active subset of those features is selected, combined, and read out. Polysemy is the simplest linguistic manifestation of this problem. A single lexical item, such as "bank", can carry multiple, mutually exclusive meanings depending on context. But the same geometric tension appears more generally in feature superposition, role binding, compositional semantics, and latent concept selection: many possible semantic degrees of freedom must coexist, while only some can be made epistemically accessible in a given context. We use polysemy as a controlled entry point into this broader capacity problem because it makes the contextual dependence of meaning explicit.

In lexical semantics, polysemy is distinguished from homonymy by the relatedness of senses [2]; for our purposes, the key feature is that a single surface form must access distinct semantic content in distinct contexts. Modern Transformer-based LLMs handle this by producing *dynamic* contextual embeddings: the vector representation of a token is iteratively transformed by self-attention to absorb information from surrounding tokens [3]. This approach is empirically successful but conceptually opaque, since the same surface symbol is mapped to different vectors depending on its sentential neighborhood.

The paper contains two levels of claim. The first is a general geometric claim: concentration of measure permits large semantic capacity but accumulated interference limits simultaneous accessibility. This claim is independent of the hinge formalism. The second is a specific organizational hypothesis: some contextual semantic readout in trained models may be organized around stable token-associated hinges and sense-conditioned subspaces. This second claim is not derived from concentration alone; it is an empirical hypothesis motivated by the geometry and tested by the predictions below. The intended empirical domain of the hinge hypothesis is not an unspecified exceptional subset of all tokens. It is meant to apply, if at all, to canonically polysemous tokens with established sense inventories—for example WordNet-, OntoNotes-, SemCor-, or WiC-style sense distinctions for which enough contextual instances are available. Failure on a representative sample of such tokens, against the control tests described in Section V D, would count as evidence against hinge anchoring, even if the

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broader capacity/accessibility account remains intact.

The hinge-and-frame formalism introduced below should therefore be read as a specific model class for organizing contextual semantic readout, not as a theorem forced by concentration of measure alone. The formalism has empirical content only if trained models privilege a stable token-associated direction—for example the static input embedding, an average contextual embedding, or a learned surrogate direction—relative to which contextual hidden states decompose into an engagement component and a sense-disambiguating component. This hinge-anchoring hypothesis is a central empirical assumption of the paper, and Section V D states how it can be falsified.

We propose a geometric interpretation inspired by the shared-vector structure of contextual frames. On this reading, a polysemous token can be analyzed as a single fixed unit vector—a *candidate hinge vector*—that is shared as a common member of multiple contextual frames. The hinge vector itself does not change; what changes is the contextual frame or subspace against which the surrounding linguistic state is read out. Crucially, the hinge vector plays an operational role within the analysis: it *gates* the readout by setting its overall magnitude, while the specific sense is decoded from the projection of the contextual hidden state onto the hinge-perpendicular carrier space.

We do not claim that this static formalism is a competing *architecture* to dynamic Transformer embeddings. Rather, we claim that the formalism gives a possible structural decomposition of what dynamic contextual representations may be doing. Layer activations that move "bank" toward different regions of activation space in financial versus river contexts can be analyzed by asking whether, relative to some token-associated hinge, the sense-discriminating variation lies primarily in the perpendicular component. The empirical question is then not which architecture is correct, but whether trained models exhibit hinge-anchored organization. Section V D sets out specific probes by which this can be tested.

The thesis of the paper is that *high-dimensional concentration-of-measure geometry provides the kinetic capacity required for contextual semantic encoding, but also imposes an epistemic accessibility limit on simultaneous semantic readout.* Concentration supplies the geometric mechanism: in high dimensions, almost all candidate semantic directions are nearly orthogonal, with large deviations bounded by $2\exp(-cd\varepsilon^2)$. The same small residual overlaps, however, accumulate when many semantic components are coactivated.

The analysis is conceptually adjacent to the *superposition hypothesis* in mechanistic interpretability [4], which holds that neural networks pack many features into nearly-orthogonal directions in a low-dimensional space. The present formalism differs in separating two issues. The general superposition picture concerns the overcomplete dictionary of feature directions and the sparse active sets that make such dictionaries usable. The hinge-and-frame picture adds a more specific organizational hypothesis: some contextual semantic readout may be structured around stable token-associated hinge directions, with sense information carried by low-dimensional subspaces in the orthogonal carrier.

The paper is organized as follows. Section II develops the geometric foundations: quasi-orthogonality, Lévy's lemma on the linear overlap, the relation to Johnson–Lindenstrauss, stable sense frames, exponential packing, and the kinetic–epistemic distinction. Section III introduces hinge vectors, the gated-subspace readout of context states, the shared-vector frame analogy, a Grassmannian packing statement for distinguishable low-dimensional sense subspaces sharing a hinge vector, and the relation to dynamic Transformer embeddings. Section IV discusses self-attention as soft routing rather than literal dual-frame reconstruction, and formulates frame selection as a latent-variable model. Section V addresses the relation to the superposition hypothesis and sparse autoencoders, the resulting semantic capacity limit, limitations including the anisotropy threat, empirical predictions, and the scope of the quantum analogy.

Notation. Throughout, $\mathbb{R}^d$ denotes Euclidean d-space with the standard inner product $\boldsymbol{u}^\intercal \boldsymbol{v}$ and induced norm $\|\boldsymbol{v}\| = \sqrt{\boldsymbol{v}^\intercal \boldsymbol{v}}$; $S^{d-1} = \{\boldsymbol{v} \in \mathbb{R}^d : \|\boldsymbol{v}\| = 1\}$ is the unit sphere; $\boldsymbol{I}$ is the identity matrix; $(\cdot)^\intercal$ denotes transpose; $\|\cdot\|_{\mathrm{op}}$ is the operator (spectral) norm; $A \succeq 0$ means A is positive semi-definite; $\mathbb{P}$ and $\mathbb{E}$ denote probability and expectation under the uniform (Haar) measure on S^{d-1} unless otherwise noted; $O(\cdot)$ and $\Omega(\cdot)$ are the standard Landau symbols. We write $\mathrm{ReLU}(x) = \max(0, x)$ and use σ for the logistic sigmoid $\sigma(x) = (1 + e^{-x})^{-1}$. Universal positive constants are denoted c, c', possibly differing between occurrences. We write d_{eff} for the effective dimension that controls concentration in the representation actually being probed; in an ideal isotropic representation $d_{\mathrm{eff}} = d$.

II. CONCENTRATION GEOMETRY AND FRAME CAPACITY

A. Quasi-Orthogonality

Let $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_N \in \mathbb{R}^d$ be unit vectors. The vectors are *strictly orthogonal* if $\boldsymbol{v}_i^\intercal \boldsymbol{v}_j = 0$ for all $i \neq j$. By elementary linear algebra, at most $N = d$ such vectors can coexist in $\mathbb{R}^d$.

This linear bound is restrictive. Many semantic features that one might wish to encode as directions need not be perfectly distinguishable; it suffices that they be *nearly* distinguishable. We therefore introduce ε-quasi-orthogonality: a set of unit vectors is ε-quasi-orthogonal if

$$|\boldsymbol{v}_i^\intercal \boldsymbol{v}_j| \leq \varepsilon, \qquad i \neq j. \tag{1}$$

Since exact orthogonality is neither achievable nor necessary in realistic embedding spaces, quasi-orthogonality is the operationally meaningful notion throughout this paper.

B. Lévy's Lemma on the Linear Overlap

The exponential capacity of high-dimensional spaces for quasi-orthogonal vectors is closely related to concentration-of-measure phenomena. In high dimensions, most of the volume of the unit sphere S^{d-1} lies near the equator relative to any fixed direction. Consequently, two independently sampled random unit vectors are almost orthogonal with overwhelming probability.

This may be formalized using Lévy's lemma [5, 6]. Let $f\colon S^{d-1} \to \mathbb{R}$ be an L-Lipschitz function, i.e., $|f(\boldsymbol{x}) - f(\boldsymbol{y})| \le L\|\boldsymbol{x} - \boldsymbol{y}\|$ for all $\boldsymbol{x}, \boldsymbol{y} \in S^{d-1}$. Then for any $\delta > 0$,

$$\mathbb{P}(|f(\boldsymbol{v}) - \mathbb{E}[f]| \ge \delta) \le 2\exp\left(-\frac{c\,d\,\delta^2}{L^2}\right), \tag{2}$$

where $c > 0$ is a universal constant and the probability is taken over the uniform measure on S^{d-1}.

To obtain quasi-orthogonality, we apply Lévy's lemma to the *linear* overlap $f_{\boldsymbol{u}}(\boldsymbol{v}) = \boldsymbol{u}^\intercal \boldsymbol{v}$, which is 1-Lipschitz by the Cauchy–Schwarz inequality. By symmetry, $\mathbb{E}[f_{\boldsymbol{u}}] = 0$. Therefore, for any $\varepsilon > 0$,

$$\mathbb{P}(|\boldsymbol{u}^\intercal \boldsymbol{v}| \ge \varepsilon) \le 2\exp(-c\,d\,\varepsilon^2). \tag{3}$$

For the standard normalized surface measure on S^{d-1}, an explicit form of the constant gives $2\exp(-(d-1)\varepsilon^2/2)$ [5]. Two independent random unit vectors are therefore ε-quasi-orthogonal with probability at least $1 - 2\exp(-cd\varepsilon^2)$, and typical overlap amplitudes scale as $d^{-1/2}$.

In ideal isotropic geometry the ambient dimension is d. In trained representations, however, anisotropy and low intrinsic dimension may reduce the effective dimension. We therefore use d_{eff} for the dimension that controls concentration in the representation actually being probed. In the isotropic idealization, $d_{\text{eff}} = d$; after whitening or restriction to the participation subspace, d_{eff} should be interpreted as the corresponding effective or intrinsic dimension.

The geometric significance is immediate: in sufficiently high effective dimension, almost all directions are nearly orthogonal to any given direction, and this quasi-orthogonality is a robust, generic property of high-dimensional spheres.

C. Relation to the Johnson–Lindenstrauss Lemma

The concentration phenomenon underlying Eq. (3) is closely related to the Johnson–Lindenstrauss (JL) lemma [7], which states that any set of n points in Euclidean space can be projected into a space of dimension $k = O(\varepsilon^{-2}\log n)$ while preserving all pairwise distances to within a factor of $(1 \pm \varepsilon)$.

Conceptually, both results arise from the same high-dimensional geometric fact: random projections and random directions are highly regular in large dimension. The JL lemma emphasizes the stability of finite metric structure under random projection, whereas Lévy's lemma emphasizes the concentration of Lipschitz observables around their mean values on high-dimensional spheres. The two are mathematically intertwined—one can derive the JL lemma from concentration of measure on the sphere applied to projection norms [8]—but they highlight complementary aspects of the same phenomenon.

For the present purposes, Lévy's lemma is the more natural tool, because the relevant issue is not dimensionality reduction of a finite dataset, but the typical quasi-orthogonality and abundance of semantic directions in high-dimensional embedding spaces.

D. Stable Sense Frames

A contextual frame used for readout must be well-conditioned. Pairwise bounds such as $|\boldsymbol{e}_i^\intercal \boldsymbol{e}_j| \le \varepsilon$ do not guarantee a well-conditioned frame when many vectors are involved, because the cumulative spectral effect of small off-diagonal entries can be significant. By the Gershgorin circle theorem, pairwise quasi-orthogonality only implies $\|\boldsymbol{G} - \boldsymbol{I}\|_{\text{op}} \le (n-1)\varepsilon$ for an n-element frame, which is uninformative when n is large. We therefore require a stronger condition on the frame actually used for readout.

Definition 1 (ρ-stable sense frame)**.** *Let $\boldsymbol{v}_0 \in S^{d-1}$ be a hinge vector and let $\mathcal{S}^{\mathcal{C}} \subset \boldsymbol{v}_0^\perp$ be a $q_{\mathcal{C}}$-dimensional sense subspace with frame matrix*

$$\boldsymbol{F}_{\mathcal{S}}^{\mathcal{C}} = [\boldsymbol{e}_2^{\mathcal{C}} \ \cdots \ \boldsymbol{e}_{q_{\mathcal{C}}+1}^{\mathcal{C}}].$$

*The sense frame is called ρ-*stable *if its Gram matrix*

$$\boldsymbol{G}_{\mathcal{S}}^{\mathcal{C}} = (\boldsymbol{F}_{\mathcal{S}}^{\mathcal{C}})^\intercal \boldsymbol{F}_{\mathcal{S}}^{\mathcal{C}}$$

satisfies

$$\|\boldsymbol{G}_{\mathcal{S}}^{\mathcal{C}} - \boldsymbol{I}\|_{\text{op}} \le \rho < 1. \tag{4}$$

Exact orthonormal sense frames correspond to $\rho = 0$.

Stability ensures that all singular values of $\boldsymbol{F}_{\mathcal{S}}^{\mathcal{C}}$ lie between $\sqrt{1-\rho}$ and $\sqrt{1+\rho}$. Consequently, the dual sense frame

$$\widetilde{\boldsymbol{F}}_{\mathcal{S}}^{\mathcal{C}} = \boldsymbol{F}_{\mathcal{S}}^{\mathcal{C}}(\boldsymbol{G}_{\mathcal{S}}^{\mathcal{C}})^{-1}$$

is well-defined on $\mathcal{S}^{\mathcal{C}}$ and satisfies

$$\|\widetilde{\boldsymbol{F}}_{\mathcal{S}}^{\mathcal{C}}\|_{\text{op}} \le (1-\rho)^{-1/2}.$$

The dual-frame readout used below is therefore numerically stable provided the selected sense frame is stable. No conditioning assumption involving the hinge direction itself is required, since the readout is performed after projection onto $\boldsymbol{v}_0^\perp$.

E. Exponential Packing

The concentration bound (3) yields an exponential packing result for individual vectors. Sample N unit vectors independently and uniformly from S^{d-1}. By a union bound over the $\binom{N}{2} < N^2/2$ pairs,

$$\mathbb{P}(\exists\, i \neq j : |\boldsymbol{v}_i^\intercal \boldsymbol{v}_j| \geq \varepsilon) < N^2 \exp(-c\, d_{\text{eff}}\, \varepsilon^2). \tag{5}$$

This probability is less than one—and hence an ε-quasi-orthogonal family of size N exists—provided

$$N < \exp\left(\tfrac{c}{2}\, d_{\text{eff}}\, \varepsilon^2\right). \tag{6}$$

For typical LLM embedding dimensions ($d \sim 10^3$ to 10^4) and modest tolerances ($\varepsilon \sim 0.1$), the nominal bound permits enormous numbers of quasi-orthogonal directions. In applications, however, the relevant quantity is the effective dimension d_{eff} of the representation being probed, not necessarily the raw ambient dimension. We refer to the resulting supply of approximately independent directions as the *quasi-dimensionality* of the representation.

We emphasize, however, that the packing bound controls only pairwise overlaps. As the Gershgorin remark above shows, pairwise control does not by itself yield operator-norm control over a large frame matrix. The frame-level analogue requires separate conditioning assumptions or constructions.

F. Kinetic Capacity Versus Epistemic Accessibility

The exponential packing bound expresses a *kinetic* capacity: it counts what can geometrically coexist. It does not by itself determine what can be reliably read out when several semantic components are simultaneously active. This distinction is essential for interpreting concentration geometry in LLMs.

Let N candidate semantic features be represented by unit directions $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_N \in \mathbb{R}^{d_{\text{eff}}}$, or by directions in an effective d_{eff}-dimensional subspace of a larger ambient activation space. Suppose that, in a given context, a subset $S \subset \{1, \ldots, N\}$ of size $|S| = k$ is active, with amplitudes x_j, producing the hidden state

$$\boldsymbol{h} = \sum_{j \in S} x_j \boldsymbol{w}_j. \tag{7}$$

A linear readout of feature i is

$$\boldsymbol{w}_i^\intercal \boldsymbol{h} = x_i + \sum_{\substack{j \in S \\ j \neq i}} x_j\, \boldsymbol{w}_i^\intercal \boldsymbol{w}_j. \tag{8}$$

The first term is the signal; the second is interference.

For random incoherent directions, concentration of measure gives

$$\mathbb{E}\big[\boldsymbol{w}_i^\intercal \boldsymbol{w}_j\big] = 0, \qquad \mathrm{Var}\big(\boldsymbol{w}_i^\intercal \boldsymbol{w}_j\big) \simeq \frac{1}{d_{\text{eff}}}. \tag{9}$$

Consequently, conditional on the active amplitudes,

$$\mathrm{Var}\left(\sum_{\substack{j \in S \\ j \neq i}} x_j\, \boldsymbol{w}_i^\intercal \boldsymbol{w}_j\right) \simeq \frac{1}{d_{\text{eff}}} \sum_{\substack{j \in S \\ j \neq i}} x_j^2. \tag{10}$$

For amplitudes of order unity, the interference standard deviation is therefore

$$\sigma_{\text{int}} \sim \sqrt{\frac{k}{d_{\text{eff}}}}. \tag{11}$$

Equation (11) is the epistemic counterpart of the packing bound (6). Lévy concentration allows exponentially many potential semantic directions to coexist, but the residual overlaps among them are typically of order $d_{\text{eff}}^{-1/2}$, not zero. When many features are coactivated, these small overlaps accumulate. Reliable readout therefore requires an SNR condition, schematically

$$\sqrt{\frac{k}{d_{\text{eff}}}} \ll \Delta, \tag{12}$$

where Δ is the relevant decision margin determined by the feature amplitudes, nonlinear thresholds, downstream probes, and task tolerance.

Thus high-dimensional concentration gives both a blessing and a limitation:

$$\text{kinetic capacity:} \qquad N \text{ may be exponentially large in } d_{\text{eff}}, \tag{13}$$

$$\text{epistemic accessibility:} \quad k \text{ must remain small enough for bounded i} \tag{14}$$

In this sense, the embedding or residual-stream dimension acts as a *semantic concurrency bandwidth*. It does not merely limit how many meanings can be stored or geometrically hosted; it limits how many independently recoverable semantic components can be jointly active in a single finite-dimensional channel.

This also clarifies the relation to compressed sensing. In standard sparse-recovery settings, stable recovery of a k-sparse vector in an N-dimensional feature space from m measurements requires bounds of the form

$$m \gtrsim C\, k \log(N/k), \tag{15}$$

under suitable incoherence or restricted-isometry assumptions [9–11]. Here m is the number of effective independent measurements, not automatically the intrinsic dimension of a representation. In an idealized isotropic residual stream used directly as the measurement channel one may identify m with d_{eff}, but in trained models the effective representation dimension and the effective readout dimension may diverge. The compressed-sensing analogy is therefore diagnostic rather than literal: it separates the dependence on total feature vocabulary N from the dependence on active set size k.

The hinge-and-frame formalism adds a further structural hypothesis to this generic sparse-recovery picture, namely that for some tokens or concepts the relevant active features are organized around a stable hinge direction and read through sense-conditioned subspaces.

III. HINGE VECTORS AND SENSE SUBSPACES

A. The Hinge-Vector Formalism

We now introduce the hinge-and-frame formalism as a specific organizational hypothesis for contextual semantic readout. Each token t is associated with a candidate unit vector $\boldsymbol{v}(t) \in \mathbb{R}^d$, called a *hinge vector*. In an empirical model this direction might be the static input embedding, the average contextual embedding, or a learned token-specific probe direction. The formalism has predictive content only if some such direction is privileged by the trained representation.

Each context $\mathcal{C}$ in which t appears is represented by a sense frame whose first vector is identified with the hinge:

$$\boldsymbol{e}_1^{\mathcal{C}} = \boldsymbol{v}(t). \tag{16}$$

The orthogonal complement $\boldsymbol{v}(t)^{\perp}$ is the *sense carrier space* associated with the hinge. A particular context $\mathcal{C}$ selects within this carrier a $q_{\mathcal{C}}$-dimensional *sense subspace*, with $1 \le q_{\mathcal{C}} \le d-1$, spanned by unit vectors $\{\boldsymbol{e}_2^{\mathcal{C}}, \ldots, \boldsymbol{e}_{q_{\mathcal{C}}+1}^{\mathcal{C}}\}$:

$$\mathcal{S}^{\mathcal{C}} = \operatorname{span}\{\boldsymbol{e}_2^{\mathcal{C}}, \ldots, \boldsymbol{e}_{q_{\mathcal{C}}+1}^{\mathcal{C}}\} \subset \boldsymbol{v}(t)^{\perp}. \tag{17}$$

A token is *polysemous* if its hinge vector participates in two or more such contextual frames, each carrying a distinct low-dimensional sense subspace. For example, $\boldsymbol{v}(\text{bank})$ might serve as the hinge of a finance frame, a river frame, and a tilt-of-aircraft frame, each with its own low-dimensional sense subspace inside the common carrier $\boldsymbol{v}(\text{bank})^{\perp}$.

The assumption $q_{\mathcal{C}} \ll d_{\text{eff}}$ is not a theorem but an empirical modeling hypothesis. It expresses the idea that an individual sense constrains only a small number of semantic degrees of freedom relative to the full activation space. A word sense is not a single direction: it may involve topic, syntax, selectional preferences, discourse role, and world-knowledge associations. But it is also not expected to occupy the full residual stream. The low-dimensional subspace assumption is therefore an intermediate claim: senses are extended regions, yet their dominant variation is low rank. This is precisely what the operational subspace test in Section V D examines by estimating the number of principal directions needed to explain within-sense variance.

The gated-subspace readout of context. The meaning of a polysemous token is not contained in its hinge vector alone. Meaning is relational. It emerges from the interaction between the candidate hinge vector and the surrounding linguistic context, mediated by the selected sense subspace.

Given an input context x, let $\boldsymbol{h}(x) \in \mathbb{R}^d$ denote the contextual state—in a static model, a vector derived from the surrounding tokens; in a trained model, the hidden state produced by preceding layers. We decompose $\boldsymbol{h}(x)$ along the hinge:

$$\boldsymbol{h}(x) = h_{\|}(x)\,\boldsymbol{v}(t) + \boldsymbol{h}_{\perp}(x), \qquad h_{\|}(x) = \boldsymbol{v}(t)^{\intercal}\boldsymbol{h}(x), \tag{18}$$

with $\boldsymbol{h}_{\perp}(x) \in \boldsymbol{v}(t)^{\perp}$. The two components play distinct roles: $h_{\|}(x)$ measures how strongly the context *engages* the token-associated direction, while $\boldsymbol{h}_{\perp}(x)$ carries the *sense-disambiguating* information.

The dual-frame readout in frame $\mathcal{C}$ is performed within the sense subspace. Let $\boldsymbol{F}_{\mathcal{S}}^{\mathcal{C}} = [\boldsymbol{e}_2^{\mathcal{C}} \;\cdots\; \boldsymbol{e}_{q_{\mathcal{C}}+1}^{\mathcal{C}}]$ be the frame matrix of the selected sense subspace, and let $\widetilde{\boldsymbol{F}}_{\mathcal{S}}^{\mathcal{C}}$ denote its dual. Then

$$\boldsymbol{\alpha}^{\mathcal{C}}(x) = \big(\widetilde{\boldsymbol{F}}_{\mathcal{S}}^{\mathcal{C}}\big)^{\intercal}\,\boldsymbol{h}_{\perp}(x) \in \mathbb{R}^{q_{\mathcal{C}}} \tag{19}$$

gives the dual-frame coefficients of $\boldsymbol{h}_{\perp}(x)$ in $\mathcal{S}^{\mathcal{C}}$.

The *sense score* of frame $\mathcal{C}$ for context x is modeled as

$$s_{\mathcal{C}}(x) = g\big(h_{\|}(x)\big) \cdot r_{\mathcal{C}}\big(\boldsymbol{h}_{\perp}(x)\big), \tag{20}$$

where $g\colon \mathbb{R} \to \mathbb{R}_{\ge 0}$ is an engagement gate and $r_{\mathcal{C}}\colon \boldsymbol{v}(t)^{\perp} \to \mathbb{R}_{\ge 0}$ is a within-carrier selector: it scores how well the sense subspace $\mathcal{S}^{\mathcal{C}}$ aligns with $\boldsymbol{h}_{\perp}(x)$. Three natural choices for $r_{\mathcal{C}}$ are

1. the squared norm of the dual-frame expansion, $r_{\mathcal{C}} = \|\boldsymbol{\alpha}^{\mathcal{C}}(x)\|^2$, when the subspace is characterized purely by its orientation;

2. the squared overlap with a designated direction $\boldsymbol{w}^{\mathcal{C}} \in \mathcal{S}^{\mathcal{C}}$, $r_{\mathcal{C}} = (\boldsymbol{w}^{\mathcal{C}\intercal}\boldsymbol{h}_{\perp}(x))^2$, when a single sense-distinguishing axis is appropriate; or

3. a learned positive functional, e.g. a quadratic form $\boldsymbol{h}_{\perp}(x)^{\intercal}A^{\mathcal{C}}\boldsymbol{h}_{\perp}(x)$ with $A^{\mathcal{C}} \succeq 0$, when training infers the relevant within-carrier structure.

The selected frame is

$$\mathcal{C}^*(x) = \arg\max_{\mathcal{C}} s_{\mathcal{C}}(x). \tag{21}$$

The meaning of token t in context x, within this model class, is the relational triple

$$\big(\boldsymbol{v}(t),\; \mathcal{C}^*(x),\; \boldsymbol{\alpha}^{\mathcal{C}^*(x)}(x)\big). \tag{22}$$

On the form of the gating prefactor. The score (20) is a deliberately simple factorized model of contextual readout. It separates two functions: a scalar engagement gate $g(h_{\|})$ and a within-carrier selector $r_{\mathcal{C}}(\boldsymbol{h}_{\perp})$. The minimal constraints on the gate are

$$g(0) = 0, \qquad g(h_{\|}) \ge 0, \qquad g \text{ nondecreasing in } |h_{\|}|.$$

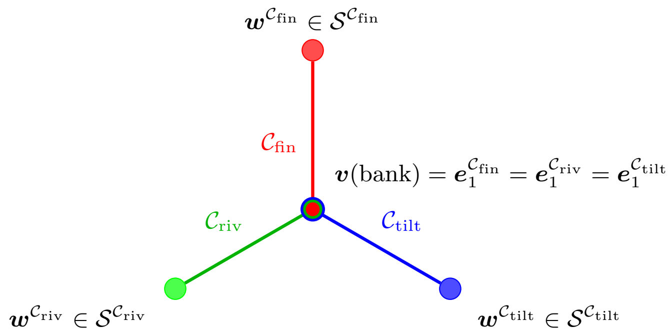

FIG. 1. Schematic of hinge-anchored frame sharing. A candidate hinge vector associated with the token "bank" is shared by several possible sense frames. The colored outer nodes represent representative sense directions, not entire subspaces. The formal model treats sense content as lying in low-dimensional subspaces of the hinge-perpendicular carrier $\boldsymbol{v}(\text{bank})^{\perp}$.

The choice $g(h_\parallel) = h_\parallel^2$ is convenient because it is smooth and sign-insensitive; other gates, such as $|h_\parallel|$, $\text{ReLU}(h_\parallel)$, or learned sigmoid gates, are possible.

The multiplicative form $g(h_\parallel) r_{\mathcal{C}}(\boldsymbol{h}_\perp)$ should not be interpreted as uniquely forced. It is one parsimonious way of encoding the logical requirement that a sense should score highly only when the token is engaged and the perpendicular context fits that sense. Other t-norm-like combinations could implement a similar conjunction. The empirical content of the formalism is therefore not the precise choice of g, but the predicted separation between an engagement coordinate parallel to the hinge and sense-disambiguating structure inside the hinge-perpendicular carrier. This is what the probes in Section V D test.

Why the hinge vector matters. In this formalism, the hinge $\boldsymbol{v}(t)$ plays an operational role through the decomposition (18). When the contextual state is orthogonal to $\boldsymbol{v}(t)$, no frame associated with t scores positively if $g(0) = 0$. When $h_\parallel(x)$ is large, the readout is amplified and the within-carrier selector $r_{\mathcal{C}}$ decides which sense applies. Different candidate hinges induce different decompositions $\boldsymbol{h} = h_\parallel \boldsymbol{v} + \boldsymbol{h}_\perp$, and hence different sense carriers $\boldsymbol{v}(t)^{\perp}$. The hinge hypothesis is therefore not mere bookkeeping: it predicts that some token-associated direction is privileged for sense disambiguation.

Figure 1 schematically illustrates a simple hinge-anchored organization: three contextual frames share a single candidate hinge vector $\boldsymbol{v}(\text{bank})$, each carrying a distinct representative sense direction or low-dimensional sense subspace inside $\boldsymbol{v}(\text{bank})^{\perp}$.

B. Shared-Vector Frame Structure and the Contextuality Analogy

The word "contextual" is used here primarily in the linguistic and geometric sense: the interpretation of a vector depends on the frame or subspace relative to which it is read. This resembles, but should not be identified with, quantum contextuality. In quantum theory, Kochen–Specker contextuality concerns the impossibility of assigning noncontextual values consistently across intertwined measurement contexts [12, 13]. The present paper does not prove, assume, or require any semantic analogue of that no-go theorem.

What we borrow is only the shared-vector frame structure: a single vector may be common to several incompatible frames, and its operational role depends on which frame is used for readout. In the semantic setting, the shared vector is a candidate hinge associated with a token, while the alternative frames or subspaces correspond to context-dependent senses or latent semantic roles.

The central geometric question is whether sufficiently many nearly independent sense directions or low-dimensional sense subspaces can coexist in $\boldsymbol{v}(t)^{\perp} \cong \mathbb{R}^{d-1}$ without destructive interference. This question has a two-sided answer:

1. shared-vector frame structure provides a possible combinatorial architecture—the logical possibility that a single hinge vector simultaneously belongs to multiple alternative frames, each defining a distinct sense subspace;

2. concentration of measure provides the kinetic capacity—the guarantee that in high dimensions, exponentially many candidate directions or low-dimensional subspaces can coexist with small pairwise coherence;

3. accumulated interference provides the epistemic limitation—the warning that coexistence does not imply simultaneous readability when many semantic components are active at once.

By Lévy's lemma applied within $\boldsymbol{v}(t)^{\perp}$, once the hinge vector is fixed, the overwhelming majority of candidate sense directions remain quasi-orthogonal to one another. A single hinge may therefore support a large number of approximately independent candidate senses. Concentration of measure supplies the geometric substrate that makes contextual encoding scalable, while the SNR condition (12) constrains how many of those candidate senses can be jointly active and reliably read out.

C. Many Sense Subspaces Sharing One Hinge Vector

The packing bound (6) counts individual quasi-orthogonal vectors. The hinge-and-frame claim concerns sense *subspaces*—structured low-dimensional subspaces of a fixed hinge-perpendicular carrier. The following proposition states the corresponding Grassmannian packing fact.

Proposition 1 (Grassmannian sense-subspace packing). *Fix a unit vector $\boldsymbol{v}_0 \in S^{d-1}$ and let $m = d-1$. Let $1 \le q \ll m$. For every fixed coherence threshold*

$$\mu > 2\sqrt{\frac{q}{m}},$$

there exists a family of q-dimensional subspaces

$$\mathcal{S}_1, \dots, \mathcal{S}_M \subset \boldsymbol{v}_0^{\perp}$$

such that

$$\|U_a^{\intercal} U_b\|_{\mathrm{op}} \le \mu, \qquad a \neq b, \tag{23}$$

where $U_a, U_b \in \mathbb{R}^{m\times q}$ are orthonormal basis matrices for $\mathcal{S}_a$ and $\mathcal{S}_b$, and

$$M \ge \exp(c_{\mu,q} m) = \exp(c_{\mu,q}(d-1)) \tag{24}$$

for a positive constant $c_{\mu,q}$.

Proof sketch. This is a standard Grassmannian packing consequence of concentration on the Stiefel and Grassmann manifolds and Grassmannian packing bounds [14, 15]. For two independent Haar-random q-planes in $\mathbb{R}^m$, represented by orthonormal matrices $U, V \in \mathbb{R}^{m\times q}$, the singular values of $U^{\intercal}V$ are the cosines of the principal angles between the subspaces. Results on the distribution of principal angles between random subspaces imply that, in the regime $q \ll m$, the largest principal correlation $\|U^{\intercal}V\|_{\mathrm{op}}$ is of order $\sqrt{q/m}$, with the natural high-probability threshold occurring at the same $2\sqrt{q/m}$ scale as in the Gaussian comparison model [16–18]. With U fixed and the Stiefel manifold $V_q(\mathbb{R}^m)$ metrized by the embedded Frobenius norm,

$$\big|\|U^{\intercal}V_1\|_{\mathrm{op}} - \|U^{\intercal}V_2\|_{\mathrm{op}}\big| \le \|U^{\intercal}(V_1 - V_2)\|_{\mathrm{op}} \le \|V_1 - V_2\|_{\mathrm{F}},$$

so the map $V \mapsto \|U^{\intercal}V\|_{\mathrm{op}}$ is 1-Lipschitz. Stiefel concentration, obtained from Haar concentration on $O(m)$ via the Riemannian submersion $O(m) \to V_q(\mathbb{R}^m)$, gives Gaussian-type tails with rate proportional to m for deviations above the typical scale. Sampling M independent subspaces and applying a union bound over the $\binom{M}{2}$ pairs yields positive probability that all pairwise coherences are at most μ, provided M is exponential in m. Hence such a packing exists. □

For typical LLM dimensions ($d \sim 10^3$–10^4), the proposition implies that a single hinge can anchor a very large family of low-dimensional candidate sense subspaces in the ideal isotropic case. For trained representations, replacing $m = d-1$ by an effective carrier dimension is an additional modeling step, not part of the Grassmannian theorem itself. It amounts to first choosing an approximately isotropic carrier subspace of dimension $m_{\mathrm{eff}} = d_{\mathrm{eff}} - 1$—for example by whitening or by restricting to the participation subspace of the activations—and then applying the same packing argument inside that carrier. Without such an isotropic effective carrier, the Haar-random subspace model need not describe the geometry of trained activations.

This is the kinetic capacity claim. It should not be confused with epistemic accessibility: if several such subspaces contribute simultaneously to a hidden state, the residual cross-coherences accumulate and are subject to the SNR constraint of Eq. (11).

D. Relation to Dynamic Contextual Embeddings

Standard Transformer-based LLMs handle polysemy through dynamic contextual embeddings. The token "bank" is initially mapped to a static embedding $\boldsymbol{v}_0(\text{bank})$, but the representation at layer ℓ,

$$\boldsymbol{h}_\ell(t \mid x),$$

depends on the surrounding context x.

The hinge-and-frame formalism should be understood as a proposed decomposition of these dynamic trajectories, not as an alternative architecture. Given a candidate hinge direction $\boldsymbol{h}_t$ associated with token t—for example the input embedding, an average contextual embedding, or a learned probe direction—each layer state admits the decomposition

$$\boldsymbol{h}_\ell(t \mid x) = a_\ell(t,x)\,\boldsymbol{h}_t + \boldsymbol{z}_\ell(t,x), \qquad a_\ell(t,x) = \boldsymbol{h}_t^{\intercal}\boldsymbol{h}_\ell(t \mid x), \qquad \boldsymbol{z}_\ell(t,x \tag{25}$$

This is the layer-indexed version of the generic decomposition (18). Thus, when $\boldsymbol{h}(x) = \boldsymbol{h}_\ell(t \mid x)$ and $\boldsymbol{v}(t) = \boldsymbol{h}_t$, the scalar $h_{\|}(x)$ of Eq. (18) is the layer-indexed coefficient $a_\ell(t,x)$.

The decomposition itself is of course always possible for any chosen direction $\boldsymbol{h}_t$; the hypothesis concerns the layerwise behavior of its components. Operationally, hinge anchoring predicts that $a_\ell(t,x)$ tracks token engagement or salience and is relatively less sense-specific, while $\boldsymbol{z}_\ell(t,x)$ becomes increasingly sense-structured across layers. In a successful hinge-anchored representation, one expects sense labels to become more linearly or subspace-separable in $\boldsymbol{h}_t^{\perp}$ at intermediate or late layers, whereas replacing $\boldsymbol{h}_t$ by matched control directions should weaken this separation. We do not require $a_\ell(t,x)$ to be constant across layers. It may vary with salience, syntactic role, or attention routing. The claim is only that the primary sense-disambiguating variation should reside in the perpendicular component rather than in the scalar engagement coordinate.

More strongly, for each sense label s, the set of vectors

$$\{\boldsymbol{z}_\ell(t,x) : x \text{ realizes sense } s\}$$

should concentrate near a low-dimensional subspace $\mathcal{S}_s \subset \boldsymbol{h}_t^{\perp}$.

This gives a concrete bridge between dynamic contextual embeddings and the static hinge formalism. The dynamic trajectory is not claimed to literally move inside a fixed pre-existing frame. Rather, the claim is that after choosing a candidate hinge, the layerwise states decompose into a parallel engagement coordinate and a perpendicular sense-disambiguating coordinate. This claim is falsifiable: if no token-associated hinge direction yields better sense separation in $\boldsymbol{h}_t^{\perp}$ than generic control directions, then the hinge-anchoring hypothesis fails even if generic superposition and the kinetic/epistemic capacity distinction remain valid.

Table I summarizes the relationship between the standard dynamic description and the hinge-decomposition hypothesis.

IV. ATTENTION, LATENT FRAMES, AND STOCHASTIC DECODING

A. Self-Attention as Soft Routing, Not Literal Frame Reconstruction

The static contextual encoding of Section III is not identical to Transformer self-attention. A standard attention head computes

$$\boldsymbol{h}' = \sum_j \alpha_j \boldsymbol{v}_j, \qquad \alpha_j = \frac{\exp(\boldsymbol{q}^{\intercal}\boldsymbol{k}_j/\sqrt{d})}{\sum_{j'} \exp(\boldsymbol{q}^{\intercal}\boldsymbol{k}_{j'}/\sqrt{d})}, \tag{26}$$

where keys and values are produced by different learned maps, typically W_K and W_V. Therefore the value vectors are not, in general, dual-frame coefficients for the key vectors, and standard attention should not be identified with literal dual-frame reconstruction.

The connection is weaker but still useful. Attention implements a soft, data-dependent routing mechanism: the query selects which contextual directions are amplified and which are suppressed. In the hinge-and-frame language, this resembles frame or subspace selection rather than frame reconstruction. A literal dual-frame interpretation would require additional structure, for example keys forming an approximately Parseval frame and values being constrained to implement the corresponding dual reconstruction. We do not assume that trained Transformers satisfy this condition.

Thus the empirical question is not whether attention is mathematically identical to the dual-frame readout of Eq. (19). It is whether trained attention and MLP layers learn routing operations that approximate the functional role of frame selection: increasing accessibility of the context-relevant semantic components while suppressing irrelevant ones.

B. Latent-Variable Frame Selection

Frame selection may be represented as a latent-variable decomposition:

$$p(t \mid x) = \sum_{\mathcal{C}} p(t \mid x, \mathcal{C})\, p(\mathcal{C} \mid x). \tag{27}$$

Here $p(\mathcal{C} \mid x)$ is a learned distribution over contextual frames, and $p(t \mid x, \mathcal{C})$ is the token distribution obtained after reading the context state $\boldsymbol{h}(x)$ in frame $\mathcal{C}$.

Temperature may be introduced at the frame level,

$$p(\mathcal{C} \mid x) = \frac{\exp\bigl(s_{\mathcal{C}}(x)/T\bigr)}{\sum_{\mathcal{C}'} \exp\bigl(s_{\mathcal{C}'}(x)/T\bigr)}, \tag{28}$$

where $s_{\mathcal{C}}(x)$ is the gated sense score (20), or at the token level. Low temperature concentrates probability on the single frame whose sense subspace is most aligned with $\boldsymbol{h}_{\perp}(x)$ and whose hinge has nontrivial overlap with $\boldsymbol{h}(x)$; high temperature spreads probability across multiple frames, permitting ambiguous or creative generation.

We note that the latent-variable formulation does not implicitly combine frames into a single composite frame matrix; mixing happens at the level of the probability distribution over $\mathcal{C}$, not at the level of vector concatenation. Cross-frame operator-norm control of the kind discussed in Section II D is therefore not required to make Eqs. (27) and (28) well-defined. It would become relevant only if a downstream computation explicitly read out from a stack of multiple frames simultaneously—a structure not assumed here.

The frame-level temperature in Eq. (28) should not be confused with the token-level temperature used in standard decoding,

$$p(t) = \frac{\exp(z_t/T)}{\sum_{t'} \exp(z_{t'}/T)}.$$

Standard LLMs apply temperature to token logits, not to an explicit distribution over latent frames. Nevertheless, if token logits are partly mediated by latent frame selection, token-level temperature can indirectly modulate ambiguity between frame-compatible continuations. We use the frame-temperature expression only as a conceptual model of ambiguity management, not as a claim about the literal implementation of sampling in current Transformers.

V. DISCUSSION

We have proposed a geometric account of contextual semantic capacity in LLMs. The general claim is that high-dimensional concentration permits many semantic directions or subspaces to coexist, while accumulated interference limits how many can be simultaneously read out. The more specific hinge-and-frame hypothesis is that, for at least some canonically polysemous tokens or

TABLE I. Two descriptions of contextual semantic representation. The hinge-and-frame analysis is presented as a possible decomposition of dynamic representations, not as a competing architecture.

	Dynamic description	**Hinge-decomposition hypothesis**
Token state	Changes per context and layer	Decomposed relative to candidate hinge
Context carrier	Layer activations	Hinge-perpendicular component
Polysemy mechanism	Vector deformation	Gated subspace readout
Geometric resource	Network depth and width	Quasi-dimensionality
Readout	Final-layer projection	Sense-frame coefficients or probes
Limit	Activation bottlenecks	Interference/accessibility bound

latent concepts, contextual readout is organized around stable token-associated directions and sense-conditioned subspaces.

A. Relation to the Superposition Hypothesis

The proposal is closely related to the *superposition hypothesis* in mechanistic interpretability [4, 19], which holds that neural networks represent more features than they have neurons by packing those features into nearly-orthogonal directions. The capacity bound underlying superposition is precisely the concentration estimate (3) that we use here.

The superposition literature already emphasizes sparsity, interference, and nonlinear filtering; the present formulation repackages these ingredients as a capacity–accessibility distinction and specializes them to contextual semantic readout. The concentration estimate gives the kinetic side of superposition: many features may be stored in a finite-dimensional space. The interference estimate gives the epistemic side: if many superposed features are coactivated, the residual coherences add in quadrature and produce the $\sqrt{k/d_{\mathrm{eff}}}$ noise scale of Eq. (11). This is why sparsity is not an incidental assumption but a structural necessity.

The two analyses share a geometric foundation but differ structurally. The superposition hypothesis treats the embedding space as an overcomplete dictionary of feature directions, with sparsity in the activation pattern enabling many features to coexist. The hinge-and-frame analysis identifies a possible organizational pattern within such a dictionary: features or senses cluster around candidate hinge vectors, each hinge anchoring a structured family of sense subspaces. In a sparse-coding picture, polysemous tokens correspond to possible *hinge* directions: the shared engagement coordinate is common across senses, while the sense-specific variation lies in distinct subspaces of the orthogonal carrier.

Sparse autoencoders (SAEs) provide a natural empirical interface between the present theory and trained models. In the SAE picture, a layer activation is approximated by a sparse expansion over learned decoder directions, making the $\boldsymbol{w}_i$ of Eq. (7) concrete model-derived objects rather than hypothetical semantic directions. Recent work has scaled and evaluated SAEs for language-model activations, studied the geometry of SAE feature spaces, and developed automated methods for interpreting large numbers of SAE latents [20–23]. For the present proposal, SAE decoder directions are candidate feature directions for the kinetic/accessibility tests, while sense-specific clusters of SAE activations provide a way to test the stronger hinge-and-frame hypothesis.

The relation between the two views is empirical. If trained models organize their feature directions into hinge-anchored families, then the structure of senses for a polysemous token should be readable from the geometry of its sense-conditioned activations. If they do not, the kinetic/epistemic capacity distinction may still hold, but the stronger hinge-and-frame hypothesis fails.

B. A Semantic Capacity Limit

Section II F gives the central capacity distinction. Its consequence for LLMs is that model dimension should be interpreted as a concurrency bandwidth for linearly accessible semantics. A model may geometrically host far more than d_{eff} possible meanings, but a single d_{eff}-dimensional activation channel cannot make arbitrarily many of them simultaneously accessible. Depth, attention, nonlinearities, sparsity, and external memory may redistribute this burden across layers or computational steps, but they do not eliminate the underlying tradeoff for any fixed finite-dimensional superposed channel.

A refined version replaces the active-set size k by the coactivation graph

$$C_{ij} = \mathbb{P}(i, j \text{ active}) \tag{29}$$

and assigns an expected interference energy. The square appears because Eq. (8) describes the raw interference amplitude, whereas the relevant average loss or noise power is the expected square of that amplitude. If the active amplitudes have, conditional on coactivation, zero mean and pairwise uncorrelated signs, then cross terms cancel in expectation and the contribution of feature j to the interference variance in the readout of feature i is proportional to

$$\mathbb{E}[x_j^2 \mid i, j \text{ active}](\boldsymbol{w}_i^{\mathsf{T}} \boldsymbol{w}_j)^2.$$

Absorbing amplitude variance and feature importance into coefficients a_i, a_j yields

$$E_{\text{int}} = \sum_{i \neq j} C_{ij}\, a_i a_j\, (\boldsymbol{w}_i^\intercal \boldsymbol{w}_j)^2, \tag{30}$$

where a_i encodes feature amplitude or task importance.

This functional yields two immediate qualitative consequences. First, if C_{ij} is large, the contribution of the pair (i, j) to E_{int} is minimized by driving $\boldsymbol{w}_i^\intercal \boldsymbol{w}_j$ toward zero; frequently coactive features are therefore pressured toward orthogonality. Second, if C_{ij} is near zero, the geometry of the pair is nearly unconstrained by this term; rarely coactive or mutually exclusive features can share directions, be placed in the same low-dimensional factor, or become approximately antipodal if nonlinear readout makes negative interference favorable. Thus the relevant object is not merely the total number of features N, but the weighted coactivation graph C_{ij}.

A full variational theory would minimize E_{int} together with feature-benefit terms and norm or conditioning constraints. We do not develop that optimization here; Eq. (30) is used only to express the first-order geometric pressure exerted by coactivation statistics.

C. Limitations and Failure Modes

What the geometry provides is a quantitative explanation of why there is ample room for many contextual meanings to coexist with small pairwise interference; it does not by itself establish that trained models exploit that room, nor does it guarantee that all coexisting semantic possibilities are simultaneously accessible.

The framework can fail in several ways. First, training dynamics may constrain the effective embedding dimension or drive vectors into a low-dimensional submanifold. Empirical work has documented substantial *anisotropy* in trained contextual embeddings [24–26], indicating that representations occupy a narrow cone of the embedding space rather than exploiting the full quasi-dimensional capacity. More recent work has directly examined anisotropy dynamics and intrinsic dimension in Transformer representations, including layer-dependent anisotropy profiles and training-time changes in intrinsic dimension [27]. This anisotropy directly reduces the effective dimension d_{eff} in our bounds.

The threat is quantitatively serious. Estimates of the intrinsic or participation dimension of contextual embeddings may yield values far below the nominal embedding dimension. At $d_{\text{eff}} \sim 50$ and $\varepsilon = 0.1$, the exponent in our capacity bound is $c\,\varepsilon^2\, d_{\text{eff}} \sim 0.5c$, a small constant rather than a large one—meaning that the exponential capacity claim collapses to "a few" in the worst case. For the framework to apply nontrivially, the geometry that matters is not the raw embedding space but its isotropized counterpart, obtained by whitening or related preprocessing techniques [25, 26, 28]. Empirically, such methods substantially restore high-dimensional geometry and improve downstream tasks; on the present account, this is precisely because they recover the room needed for contextual encoding. We regard the anisotropy threat as the single strongest caveat to the framework, and the question of which models, layers, and preprocessing regimes admit a meaningful d_{eff} as a priority for empirical work.

Second, the gated-subspace readout requires the selected frame matrix to be well-conditioned. Near-degeneracy ($\rho \to 1$) would amplify readout noise; in practice one would need $\rho \ll 1$ for numerical stability.

Third, the analysis identifies a specific organizational pattern (hinge-and-frame) within the broader superposition picture. If trained models do not in fact organize features in this way—if, for example, polysemous tokens are represented as superpositions without any preferred hinge direction—then the analysis would be descriptively inaccurate even if its capacity bounds remain valid.

Fourth, the simple $\sqrt{k/d_{\text{eff}}}$ law assumes incoherent random-like directions and independent signs/amplitudes. Learned networks may violate these assumptions, sometimes improving on the naive bound by orthogonalizing coactive features, and sometimes performing worse through anisotropy, correlated errors, or ill-conditioned readout. The law should therefore be read as a generic scaling principle, not as a universal equality.

Fifth, the hinge-and-frame formalism contains modeling choices: the choice of hinge, the gate g, the selector $r_{\mathcal{C}}$, and the sense-subspace dimension $q_{\mathcal{C}}$. These choices must be constrained empirically. Otherwise, the formalism risks becoming a post hoc description of whatever geometry is found. The predictions below are intended precisely to prevent this: the hinge must be privileged relative to control directions, and the perpendicular component must carry sense information in a way not explained by generic clustering alone.

D. Empirical Predictions

The most important empirical issue is to separate the generic kinetic/epistemic capacity claim from the stronger hinge-anchoring hypothesis. The generic claim predicts sparse, interference-limited superposition in finite-dimensional activation spaces. The stronger claim predicts that, for canonically polysemous tokens with adequate sense-labeled data, a stable token-associated direction acts as a privileged hinge relative to which contextual states decompose into engagement and sense-disambiguating components.

The hinge hypothesis decomposes into two logically distinct claims. *Claim A* is the hinge claim: for a token t, there exists a privileged token-associated direction $\boldsymbol{h}_t$ such that sense-relevant variation is better exposed in $\boldsymbol{h}_t^\perp$ than in the orthogonal complements of matched control directions. *Claim B* is the frame claim: within $\boldsymbol{h}_t^\perp$, sense-conditioned states concentrate near low-dimensional subspaces $\mathcal{S}_s$ whose cross-sense coherences are small. Claim A can hold while Claim B fails, for instance if sense information is distributed throughout $\boldsymbol{h}_t^\perp$ without low-rank structure. Conversely, low-dimensional sense structure may exist relative to some direction that is not token-associated, in which case Claim B has an analogue but hinge anchoring fails. The predictions below therefore test these two claims separately.

The framework makes several testable predictions, all formulated against existing models and existing sense-tagged corpora (e.g., SemCor, WiC), or against controlled activation-intervention experiments.

1. *Critical hinge-anchoring test (Claim A).* For a polysemous token t, choose a candidate hinge direction $\boldsymbol{h}_t$ such as the static input embedding, mean contextual embedding, or a learned token-specific direction. Decompose layer states as in Eq. (25). The hinge-and-frame hypothesis predicts that sense labels are better separated in $\boldsymbol{h}_t^\perp$ than in orthogonal complements of random or frequency-matched control directions. The null hypothesis is that sense-conditioned contextual states cluster by sense, but no static or token-associated direction is privileged. In that case, generic superposition remains plausible, but hinge anchoring is falsified. The primary statistic may be the difference in sense-classification accuracy, or mutual information with sense labels, between the candidate hinge and the control-hinge distribution. Significance should be assessed by permutation over sense labels and by bootstrap over token instances.

2. *Operational subspace test (Claim B).* For each sense s of a token t, collect the projected states $\boldsymbol{z}_\ell(t, x) \in \boldsymbol{h}_t^\perp$ from labeled contexts and estimate a low-rank subspace by PCA or probabilistic factor analysis. Let U_s be the matrix of the first q principal directions explaining a fixed fraction of within-sense variance. The test statistic is the cross-sense coherence

$$\|U_s^\intercal U_{s'}\|_{\mathrm{op}}.$$

The null distribution should be estimated in two ways: first by a permutation test that shuffles sense labels within the token, and second by repeating the analysis with control hinges drawn from a frequency-matched set of non-target tokens or random directions with the same norm and layer statistics. The prediction is that true sense labels relative to the candidate hinge yield lower cross-sense coherence than the shuffled-label null and stronger separation than the control-hinge null.

3. *Hinge-gated readout (Claim A).* Sense-discriminating linear probes trained on the hinge-perpendicular component $\boldsymbol{h}_\perp(x)$ should perform comparably to probes trained on the full $\boldsymbol{h}(x)$, while probes restricted to the parallel component $h_\parallel(x)\boldsymbol{v}(t)$ should perform near chance on sense classification. This would directly test the operational separation between gating (parallel) and disambiguation (perpendicular).

4. *Pairwise overlap distribution.* If trained embeddings exploit the available quasi-dimensionality, the distribution of pairwise inner products $\boldsymbol{v}_i^\intercal \boldsymbol{v}_j$ among token embeddings or feature directions should be concentrated near zero with variance $\sim 1/d_{\mathrm{eff}}$. Documented anisotropy implies that $d_{\mathrm{eff}} \ll d$ in practice, providing a quantitative target for the gap.

5. *Synthetic concurrency degradation.* To isolate the geometric variable from instruction-following and attention-routing effects, construct synthetic activation interventions at a fixed layer. The directions $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_N$ should be obtained independently of the intervention data, for example as (i) decoder directions from a sparse autoencoder trained on a disjoint corpus, or (ii) linear-probe directions trained on held-out feature labels and then frozen. One then injects controlled superpositions

$$\boldsymbol{h}_k = \boldsymbol{h}_0 + \lambda \sum_{j=1}^{k} x_j \boldsymbol{w}_j$$

while varying k and holding amplitudes, layer, token position, and downstream probe fixed. Here $\boldsymbol{h}_0$ denotes a baseline activation, such as the unperturbed activation for the same prompt and token position or a matched mean activation. The prediction is smooth SNR degradation governed by an effective $\sqrt{k/d_{\mathrm{eff}}}$ scale for random-like directions, with weaker degradation for feature sets that training has orthogonalized because they frequently coactivate. Using random directions is a control for ambient geometry, whereas using SAE or probe-derived directions tests the model's learned feature organization.

6. *Coactivation-dependent geometry.* Features or senses that frequently coactivate should exhibit reduced mutual coherence relative to rarely coactive features. Conversely, mutually exclusive or anticorrelated features may be allowed larger overlap, and in some regimes may be arranged in approximately antipodal directions. This prediction operationalizes the interference-energy functional (30). The prediction is not unique to the hinge formalism, but is a useful test of the broader capacity/accessibility picture.

E. Scope of the Quantum Analogy

The analogy to quantum contextuality is structural and limited. Both formalisms involve vectors whose interpretation depends on the frame in which they are considered. However, quantum contextuality carries physical content in the form of no-go theorems such as Kochen–Specker, which constrain hidden-variable models. No analogous no-go theorem is established here for semantic embeddings.

The present framework uses only the combinatorial architecture of shared-vector frames: a candidate hinge vector may be common to several possible frames, and the relevant readout depends on which frame or subspace is selected. This is weaker than quantum contextuality in the foundational sense. The quantum analogy should therefore be read as a structural lens, not as a claim that semantic representations inherit the foundational implications of quantum mechanics.

VI. CONCLUSION

We have proposed a geometric interpretation of contextual semantic capacity in LLMs. The general thesis is that high-dimensional concentration-of-measure geometry provides the kinetic capacity required for contextual semantic encoding, while accumulated interference limits simultaneous epistemic accessibility. This part of the argument is independent of any particular architecture or hinge-based model.

The more specific hypothesis is that some contextual semantic readout may be organized around stable token-associated hinge directions. Within this analysis, the hinge vector plays an operational role: it defines a decomposition of the hidden state into a parallel engagement coordinate and a perpendicular sense-disambiguating carrier. The sense itself is decoded from a context-selected low-dimensional subspace inside that carrier. This hinge-and-frame structure is not forced by concentration geometry; it is an empirically testable model class motivated by the same geometry.

The quantitative content is summarized by the pair

$$N \lesssim \exp(c d_{\text{eff}} \varepsilon^2), \qquad \sigma_{\text{int}} \sim \sqrt{k/d_{\text{eff}}}. \tag{31}$$

The first expression describes kinetic capacity: exponentially many approximately independent semantic directions may coexist. The second describes epistemic accessibility under simultaneous activation: residual overlaps accumulate over the active set and produce an SNR-limited readout bottleneck. The first relation is an existential packing estimate under random-sampling or concentration assumptions; the second is a typical-case interference scale under random-like coactivation. They should not be read as the same kind of theorem, but as complementary scalings governing capacity and accessibility. Compressed-sensing bounds of the form $m \gtrsim k \log(N/k)$ express the same separation between the total feature vocabulary and the active recoverable subset, with m denoting the number of effective readout measurements.

We do not claim that trained Transformers must realize the hinge-and-frame structure, nor that the structure is a competitive architectural alternative to dynamic embeddings. We claim instead that the structure is a natural geometric decomposition of the contextual encoding problem, that the broader capacity/accessibility distinction is a robust consequence of high-dimensional geometry, and that whether trained models realize hinge-anchored organization is an empirically tractable question. The critical test is whether token-associated candidate hinges are privileged relative to control directions; a negative result would falsify the hinge-anchoring hypothesis while leaving the broader superposition and interference picture intact.

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